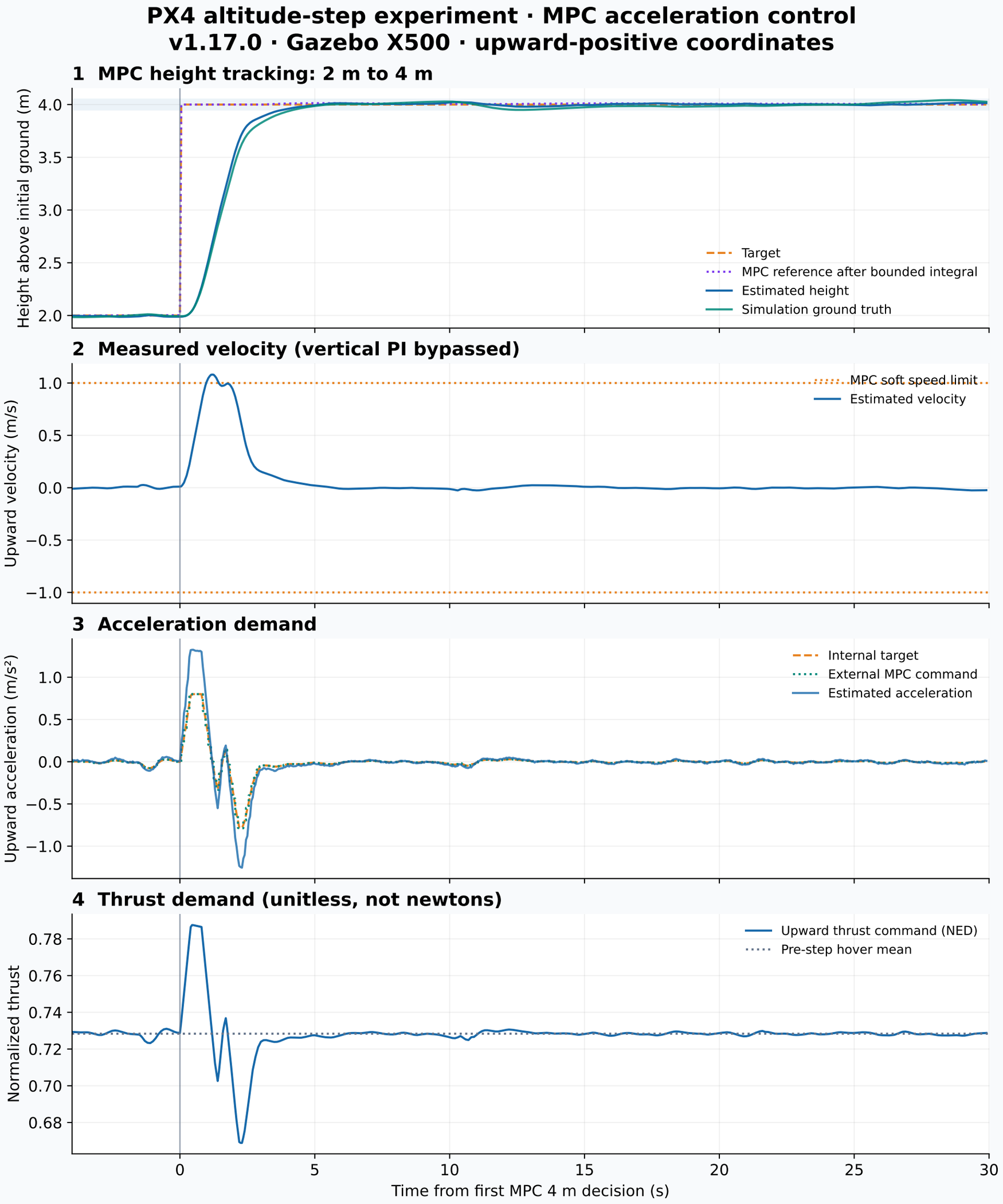
PX4 altitude-step experiment · MPC acceleration control
v1.17.0 · Gazebo X500 · upward-positive coordinates
1 MPC height tracking: 2 m to 4 m
Height above initial ground (m)
4.0
3.5
3.0
2.5
2.0
Target
MPC reference after bounded integral
Estimated height
Simulation ground truth
2 Measured velocity (vertical PI bypassed)
Upward velocity (m/s)
1.0
0.5
0.0
−0.5
−1.0
MPC soft speed limit
Estimated velocity
3 Acceleration demand
Upward acceleration (m/s²)
1.0
0.5
0.0
−0.5
−1.0
Internal target
External MPC command
Estimated acceleration
4 Thrust demand (unitless, not newtons)
Normalized thrust
0.78
0.76
0.74
0.72
0.70
0.68
Upward thrust command (NED)
Pre-step hover mean
0
5
10
15
20
25
30
Time from first MPC 4 m decision (s)